



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Sunday, 30 August 2026, 7:08 PM**State** Finished**Completed on** Sunday, 30 August 2026, 7:09 PM**Time taken** 58 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $\text{arr}[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int compare(const void *a, const void *b) {
5      long long valueA = *(const long long *)a;
6      long long valueB = *(const long long *)b;
7      if (valueA < valueB) return -1;
8      if (valueA > valueB) return 1;
9      return 0;
10 }
11
12 int main() {
13     int n;
14     if (scanf("%d", &n) != 1) {
15         return 0;
16     }
17
18     long long *arr = (long long *)malloc(n * sizeof(long long));
19     if (arr == NULL) {
20         return 1;
21     }
22
23     for (int i = 0; i < n; i++) {
24         scanf("%lld", &arr[i]);
25     }
26
27     qsort(arr, n, sizeof(long long), compare);
28
29     long long max_sum = 0;
30     for (int i = 0; i < n; i++) {
31         max_sum += arr[i] * i;
32     }
33
34     printf("%lld\n", max_sum);
35
36     free(arr);
37     return 0;
38 }
39

```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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